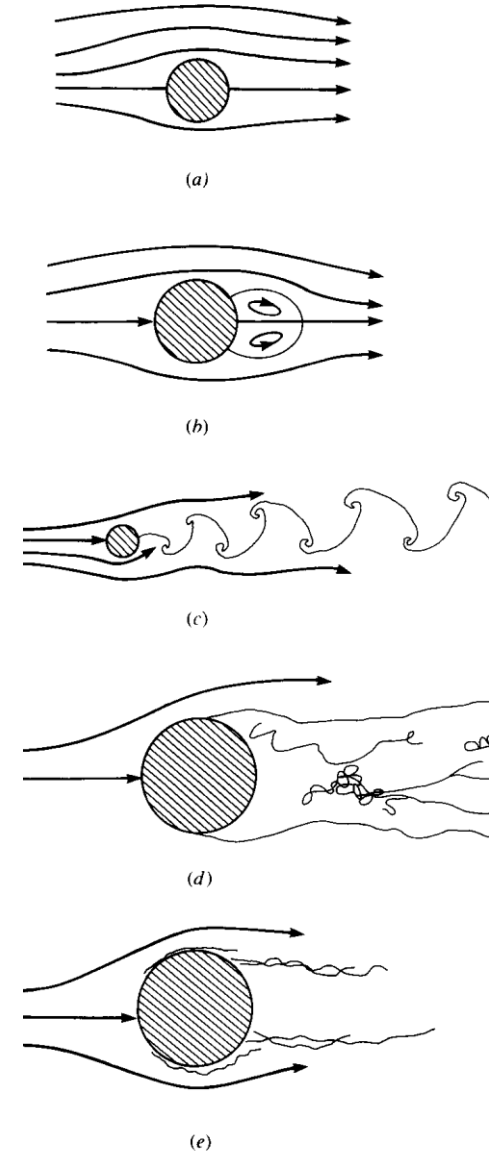
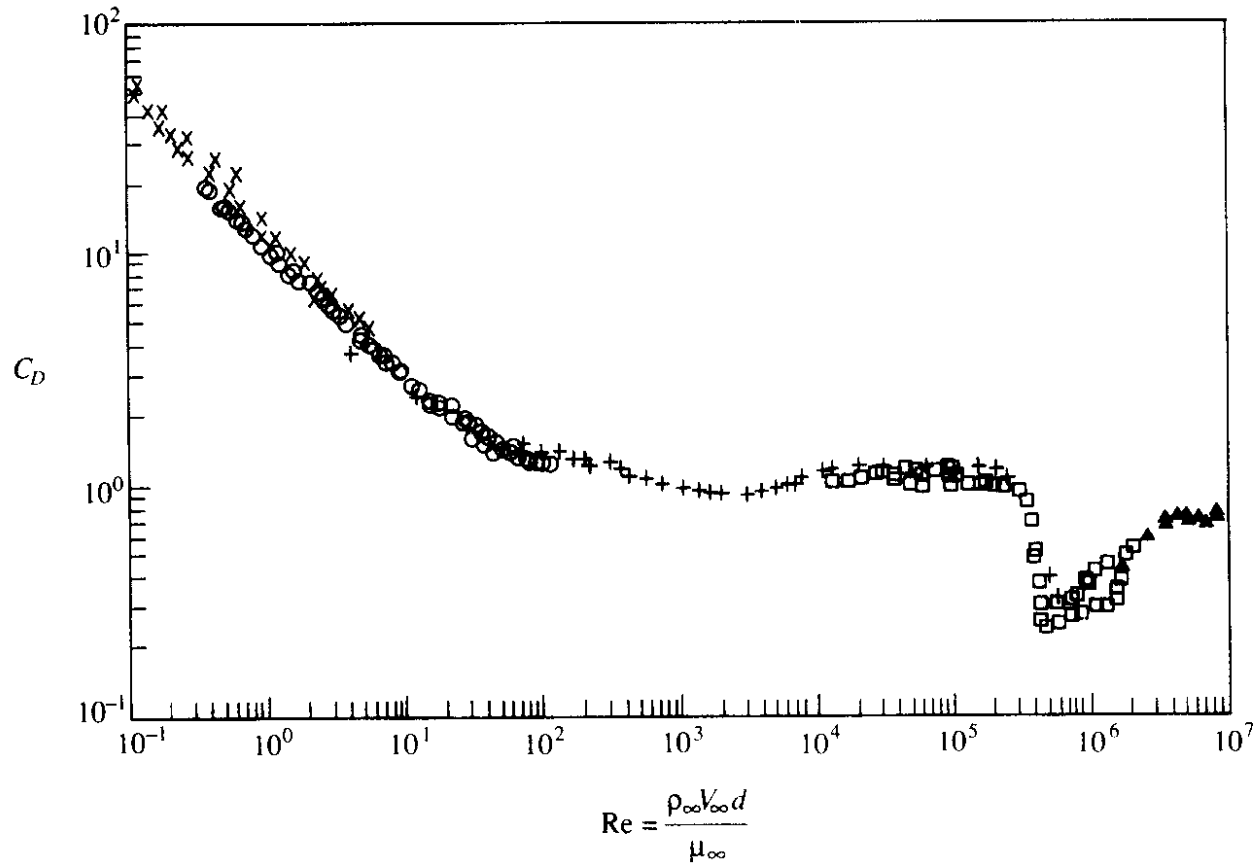


# APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE



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For very low values of  $Re$ , say,  $0 < Re < 4$ , the streamlines are almost (but not exactly) symmetrical, and the flow is attached, as sketched in Figure 3.45a. This regime of viscous flow is called *Stokes flow*; it is characterized by a near balance of pressure forces with friction forces acting on any given fluid element; the flow velocity is so low that inertia effects are very small. A photograph of this type

**$Re = O(1)$ , Only friction Drag**

For  $4 < Re < 40$ , the flow becomes separated on the back of the cylinder, forming two distinct, stable vortices that remain in the position shown in Figure 3.45b. A photograph of this type of flow is given in Figure 3.47, where  $Re = 26$ .

As  $Re$  is increased above 40, the flow behind the cylinder becomes unstable; the vortices which were in a fixed position in Figure 3.45b now are alternately shed from the body in a regular fashion and flow downstream. This flow is sketched in

*Karman vortex street,*

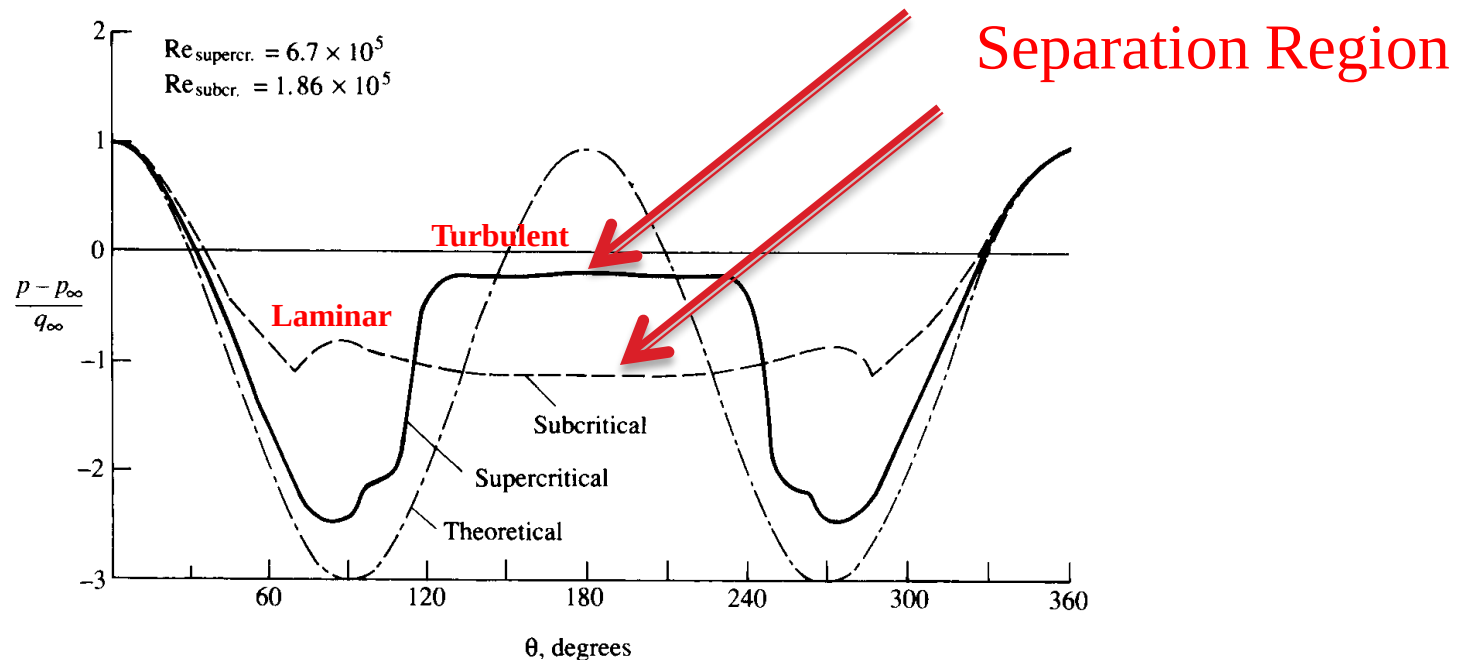
## APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE

As the Reynolds number is increased to large numbers, the Karman vortex street becomes turbulent and begins to metamorphose into a distinct wake. The laminar boundary layer on the cylinder separates from the surface on the forward face, at a point about  $80^\circ$  from the stagnation point. This is sketched in Figure 3.45*d*. The value of the Reynolds number for this flow is on the order of  $10^5$ . Note, from Figure 3.44, that  $C_D$  is a relatively constant value near unity for  $10^3 < Re < 3 \times 10^5$ .

For  $3 \times 10^5 < Re < 3 \times 10^6$ , the separation of the laminar boundary layer still takes place on the forward face of the cylinder. However, in the free shear layer over the top of the separated region, transition to turbulent flow takes place. The flow then reattaches on the back face of the cylinder, but separates again at about  $120^\circ$  around the body measured from the stagnation point. This flow is sketched in Figure 3.45*e*. This transition to turbulent flow, and the corresponding thinner wake (comparing Figure 3.45*e* with Figure 3.45*d*), reduces the pressure drag on the cylinder and is responsible for the precipitous drop in  $C_D$  at  $Re = 3 \times 10^5$

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For  $Re < 3 \times 10^6$ , the boundary layer transits directly to turbulent flow at some point on the forward face, and the boundary layer remains totally attached over the surface until it separates at an angular location slightly less than  $120^\circ$  on the back surface. For this regime of flow,  $C_D$  actually increases slightly with increasing  $Re$  because the separation points on the back surface begin to move closer to the top and bottom of the cylinder, producing a fatter wake, and hence larger pressure drag.



**Figure 3.49** Pressure distribution over a circular cylinder in low-speed flow. Comparison of the theoretical pressure distribution with two experimental pressure distributions—one for a subcritical  $Re$  and the other for a supercritical  $Re$ .